02/24/2026: Intro to Diffusion Processes

CSCI 546: Diffusion Models

Textbook reference: Chang Sec 6.1-6.4, GS Sec 13.3

Announcement (Sign-in Sheet)

Please sign the sign-in sheet.

Agenda

1. ($\approx$ 10 mins) Notes on Midterm Exam
2. ($\approx$ 10 mins) Review Problem Set #11
3. ($\approx$ 10 mins) Mini-lecture
4. ($\approx$ 25 mins) Group Exercises
5. ($\approx$ 20 mins) Briefing on Backward Equation

Review Problem Set #11

Concepts for Problem Set #12

Outline for today's material

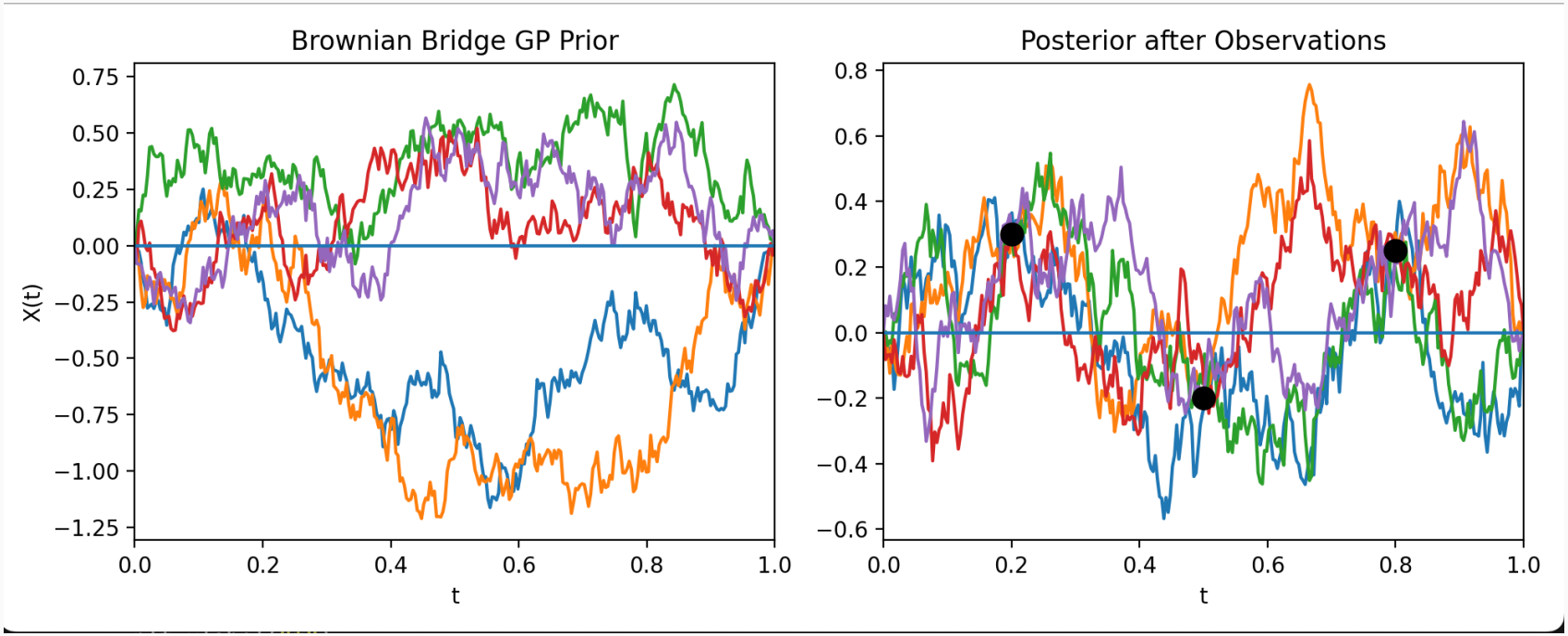
- **Brownian Bridge**
- **Diffusion processes**

Brownian Bridge

Definition. A **standard Brownian bridge** over the interval $[0, 1]$ is a standard Brownian motion $W(\cdot)$ conditioned to have $W(1) = 0$.

Brownian Bridge: Use cases

Use Case #1. *Time series modeling with constraints.* Can be used as (Gaussian process) prior over functions when endpoints are known.



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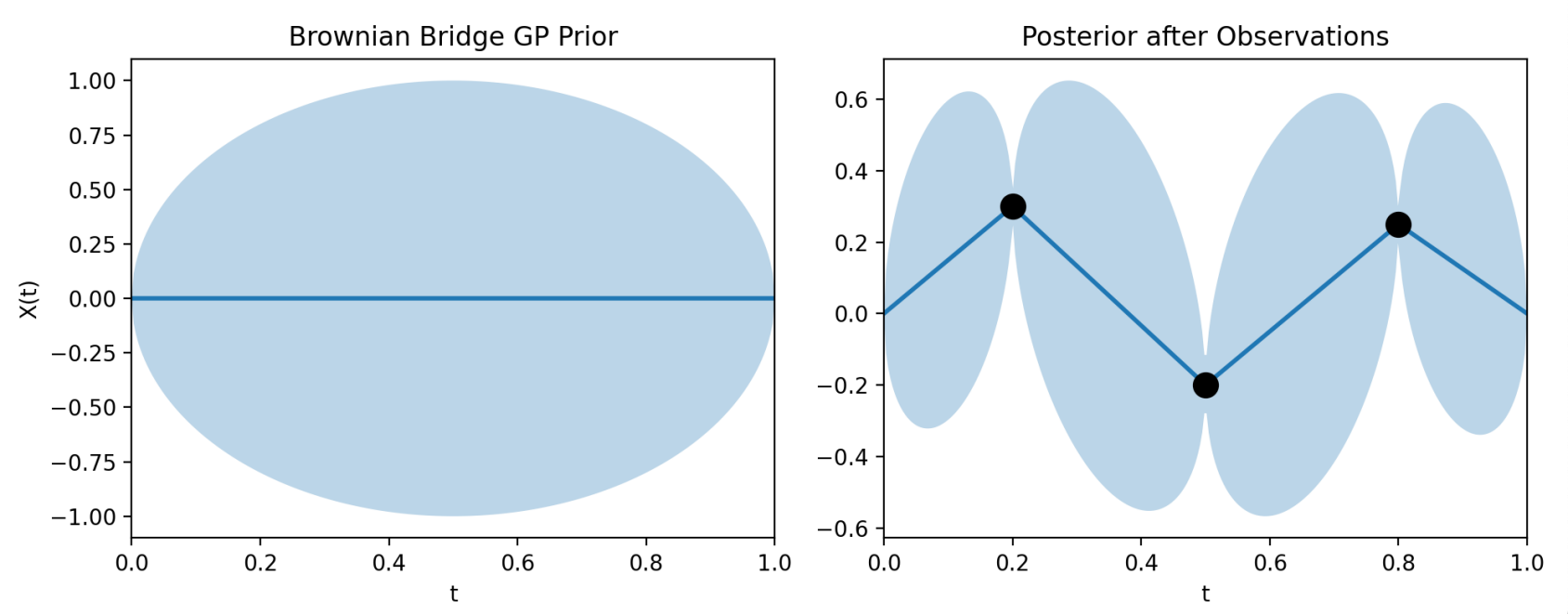

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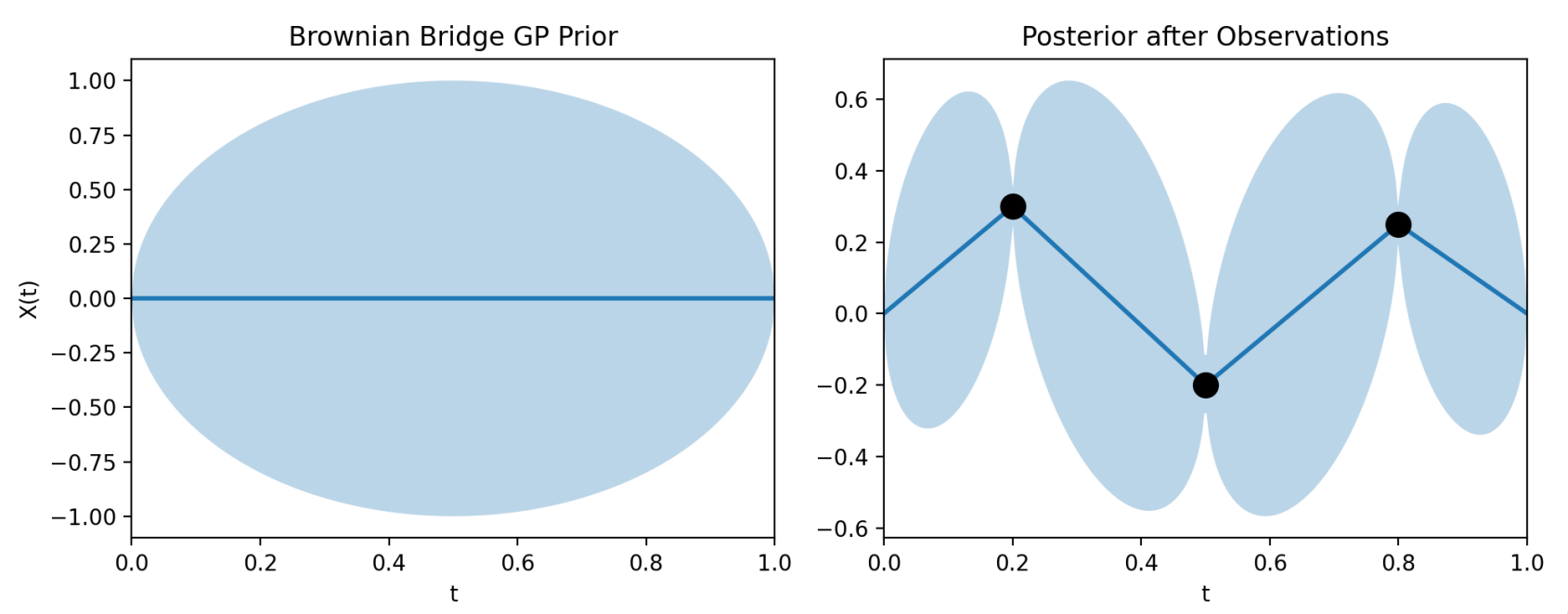
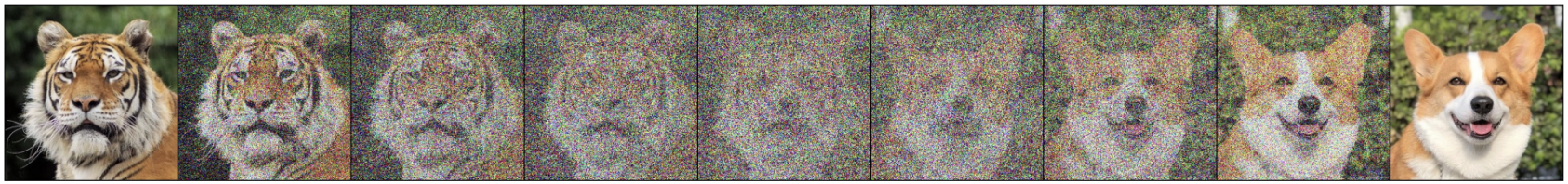


Examples:

- Financial time series with fixed start/end prices
- Sensor data with boundary constraints
- Physiological signals aligned to events.

Brownian Bridge: Use Cases

Use Case #2. *Conditional generation and interpolation.* Can be used to interpolate smoothly and stochastically between two states.



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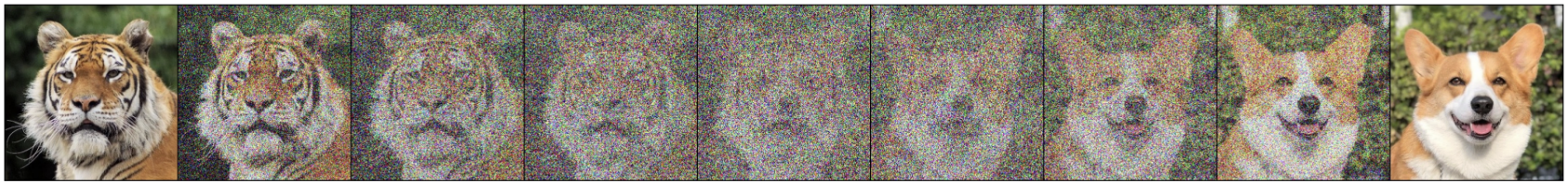

Examples:

- Image or latent-space interpolation between two images
- Motion planning with uncertainty
- Trajectory generation conditioned on start and goal.

Reference: Zhou et al (2024), Denoising bridge diffusion models, ICLR.

Outline for today's material

- Brownian Bridge
- Diffusion processes

Review: Standard Brownian Motion

Definition. A **standard Brownian motion** $\{W(t) : t \geq 0\}$ is a stochastic process having

(i) continuous paths,

(ii) stationary, independent increments, and

(iii) $W(t) \sim N(0, t)$ for all $t \geq 0$.

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(iii) $W(t) \sim N(0, t)$ for all $t \geq 0$.

Remark. The mean of a standard Brownian motion doesn't change with time, and the variance increases at rate of 1 unit per second (or whatever unit of time).

Review: Non-Standard Brownian Motion

Definition. A process X is called a (μ, σ^2) is Brownian motion if it can be written in the form

$$X(t) = X(0) + \mu t + \sigma W(t),$$

where W is a standard Brownian motion.

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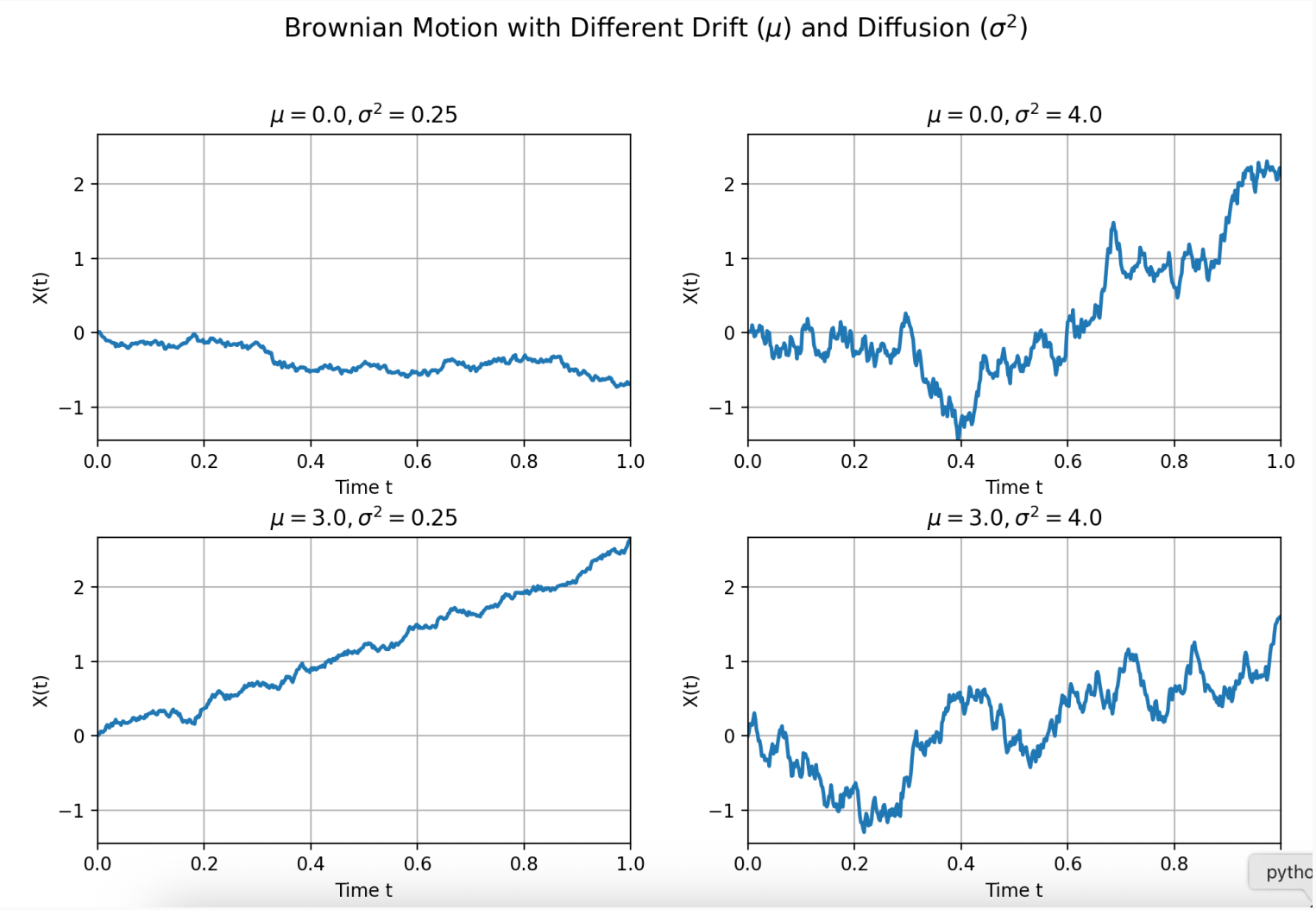
Remark. the mean and variance of a (μ, σ^2) Brownian motion increase at a rate μ and σ^2 per second (or whatever unit of time), respectively.

Review: Non-Standard Brownian Motion



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Main Ideas.

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- They are continuous-time, continuous-state Markov processes.
- You can think of them as continuous random variables that evolve over time.
- They are built up from Brownian motion.
  - We specify two functions, $\mu(\cdot)$ and $\sigma^2(\cdot)$. If the current state is X_t, then we (briefly) run a $\left(\mu(X_t), \sigma^2(X_t)\right)$ Brownian motion.
- This is analogous to how we can build up all sorts of differentiable functions (e.g. the exponential function) from lines.

A definition of a diffusion process

Definition. A process (X_t) is called a diffusion process if it has these properties

1. The time variable t is continuous.
2. It is a Markov process. (This means that X_{t_0} determines the distribution of X_t for $t > t_0$ "completely.")
3. X_t is a continuous function of t.

Reference: Johnathan Goodman, course notes.

Discretization of a diffusion process

Let $\Delta X = X_{t+\Delta t} - X_t$ be a small change in the diffusion process. Then per Chang pp.180, we have

$$\Delta X = \mu(X_t)\Delta t + \sigma(X_t)\sqrt{\Delta t}\, Z \qquad Z, \sim N(0,1)$$

Examples of Diffusion Processes

- *Brownian Motion*

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- *Brownian Bridges*

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- *Brownian Motion*
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- *Geometric Brownian Motion* - a model for stocks (the "returns", or ratios, are independent over non-overlapping intervals)
- *Ornstein-Uhlenbeck Process* - a mean reverting process
- *Wright-Fisher Process* - a process with two absorbing states (can model allele frequency in genetics)

See Chang Sec 6.2 for pictures and further details.

Random Groups

Aubrey Williams: 4
Austin Barton : 6
Blake Sigmundstad: 6
Diego Moylan: 5
Dillon Shaffer: 8
Ismoiljon Muzaffarov: 2
Jacob Tanner: 4

Josh Stoneback: 4
Joshua Bowen: 1
Joshua Culwell: 3
Laura Banaszewski: 1
Lina Hammel: 7
Logan Racz: 7

Matt Hall: 5
Micah Miller: 3
Mike Kadoshnikov: 1
Owen Cool: 2
Racquel Bowen: 3
Samuel Mocabee: 2
Tatiana Kirillova: 8

Group exercises - Problem Set 12

1. (Chang Sec 5.6) **Brownian Bridge.** Verify that the below is an easy way to manufacture a Brownian bridge from standard Brownian motion:

$$X(t) = W(t) - tW(1), \qquad \text{for } 0 \leq t \leq 1.$$

Definition. A standard Brownian Bridge is a Gaussian process X with continuous paths, mean 0, and covariance function $\mathrm{Cov}(X_s, X_t) = s(1-t)$ for $0 \leq s \leq t \leq 1$.

2. (Chang Sec 6.1) **Discretized Diffusion.** Let $\Delta X = X_{t+\Delta t} - X_t$ be a small change in the diffusion process. Then per Chang pp.180, we have

$$\Delta X = \mu(X_t)\Delta t + \sigma(X_t)\sqrt{\Delta t}\, Z \qquad Z, \sim N(0,1)$$

Show that as $\Delta t \downarrow 0$.

$$\mathbb{E}[\Delta X \mid X(t) = x] = \mu(x)\Delta t + o(\Delta t)$$
$$\mathbb{E}[(\Delta X)^2 \mid X(t) = x] = \sigma^2(x)\Delta t + o(\Delta t)$$

Definition. Let $f(x)$ and $g(x)$ be functions defined on some neighborhood of x_0. We write $f(x) = o(g(x))$ as $x \to x_0$ if

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0.$$

Equivalently, $f(x)$ becomes negligible relative to $g(x)$ as x approaches x_0.